

SENNYBRIDGE, United Kingdom

WMO: 035070

Lat: 52.0633N

Lon: 3.6147W

Elev: 1007

StdP: 14.17

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	24.9	18.7	14.6	23.9	21.6	16.8	26.6	33.5	47.5	30.5	47.0	3.9	30	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	73.5	63.5	69.9	61.4	66.9	59.6	64.7	71.2	62.6	67.5	61.1	65.0	7.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.2	86.7	66.7	60.8	82.4	64.2	59.4	78.5	62.8	30.2	71.4	28.7	67.8	27.6	65.0	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	23.3	20.3	15.6	78.9	4.1	3.5	12.7	81.4	10.3	83.5	8.0	85.5	5.1	88.0
			15.6	68.7	4.1	2.6	12.6	70.6	10.2	72.2	7.9	73.6	4.9	75.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	37.8	38.3	40.9	44.6	49.6	54.4	57.2	57.0	53.6	48.8	42.6	38.5	(6)
(7)		DBStd	8.80	6.02	5.66	5.10	5.11	5.01	4.47	4.05	4.05	4.14	5.19	5.68	6.72	(7)
(8)		HDD50	1923	379	328	282	173	67	10	1	1	12	83	228	359	(8)
(9)		HDD65	6579	844	748	746	611	476	320	245	250	342	501	673	823	(9)
(10)		CDD50	826	0	0	1	12	55	141	224	219	121	47	5	1	(10)
(11)		CDD65	7	0	0	0	0	0	1	3	3	0	0	0	0	(11)
(12)		CDH74	78	0	0	0	0	1	16	38	23	0	0	0	0	(12)
(13)		CDH80	5	0	0	0	0	0	0	3	2	0	0	0	0	(13)
(14)	Wind	WSAvg	7.8	9.3	9.2	8.3	7.6	7.6	7.0	6.3	6.3	6.7	8.4	8.0	8.6	(14)
(15)	Precipitation	PrecAvg	60.6	6.8	5.4	4.1	3.7	4.0	3.4	3.7	4.2	4.1	6.7	6.8	7.4	(15)
(16)		PrecMax	77.6	11.0	13.3	8.1	7.1	6.3	7.6	9.2	7.3	8.4	13.3	13.5	15.0	(16)
(17)		PrecMin	43.9	0.7	1.6	1.3	0.9	1.5	1.4	1.1	0.9	2.5	3.5	2.5	2.5	(17)
(18)		PrecStd	8.9	2.8	2.9	1.7	1.7	1.3	1.7	2.0	1.4	2.0	2.9	2.6	3.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.1	53.2	59.0	67.5	72.2	76.5	78.4	77.4	71.2	63.2	55.8	53.4	(19)
(20)		MCWB	50.0	49.4	50.9	55.9	61.5	65.5	65.6	65.5	63.3	59.2	54.6	52.3	52.3	(20)
(21)		2%	DB	49.5	49.4	53.7	61.1	67.0	71.0	73.9	71.6	66.4	60.0	53.9	51.5	(21)
(22)		MCWB	48.8	48.1	48.6	52.6	58.1	61.4	63.5	63.3	60.2	57.5	52.9	50.9	50.9	(22)
(23)		5%	DB	48.0	47.9	50.4	56.8	62.9	67.1	69.5	67.6	63.4	58.0	52.3	49.7	(23)
(24)		MCWB	47.3	46.8	47.5	50.0	56.0	59.2	61.1	60.8	58.4	55.9	51.2	48.9	48.9	(24)
(25)		10%	DB	46.5	46.5	48.5	53.2	59.1	63.9	66.0	64.6	61.1	56.4	50.6	48.1	(25)
(26)		MCWB	45.8	45.3	46.5	48.5	53.2	57.5	59.5	59.5	57.5	54.7	49.5	47.3	47.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.5	50.4	51.9	57.0	62.5	66.7	66.9	67.4	65.1	60.5	55.1	52.7	(27)
(28)		MCDWB	50.7	51.5	55.4	65.6	69.8	73.7	75.7	74.4	69.0	62.3	55.3	53.0	53.0	(28)
(29)		2%	WB	49.1	48.6	49.8	53.4	59.4	62.3	64.2	64.4	61.7	58.2	53.3	51.1	(29)
(30)		MCDWB	49.4	49.3	52.2	59.5	65.6	68.7	71.5	70.1	64.5	59.5	53.8	51.5	51.5	(30)
(31)		5%	WB	47.5	47.0	48.3	51.2	56.8	60.0	62.2	62.3	59.8	56.5	51.5	49.3	(31)
(32)		MCDWB	47.9	47.8	50.0	55.1	61.5	65.6	67.7	65.6	62.0	57.6	52.1	49.7	49.7	(32)
(33)		10%	WB	45.9	45.5	46.9	49.4	54.6	58.2	60.5	60.7	58.1	55.0	49.7	47.4	(33)
(34)		MCDWB	46.5	46.4	48.3	52.6	58.3	62.7	64.6	63.3	60.3	56.1	50.5	47.9	47.9	(34)
(35)	Mean Daily Temperature Range	MDBR	8.6	9.9	12.0	14.7	15.2	15.2	14.8	14.4	14.0	11.1	9.8	8.8	8.8	(35)
(36)		5% DB	MCDBR	9.0	10.7	15.8	21.4	21.3	22.4	22.7	19.5	18.0	11.7	9.3	8.9	(36)
(37)		MCWBR	9.2	8.9	11.6	13.8	13.9	14.1	13.8	12.4	12.5	9.2	8.6	9.1	9.1	(37)
(38)		5% WB	MCDBR	8.7	9.8	13.9	17.6	19.2	19.5	19.1	15.6	14.6	10.3	8.9	8.8	(38)
(39)		MCWBR	9.1	8.9	11.0	12.2	12.9	13.0	12.4	10.8	11.3	8.9	8.7	9.3	9.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.330	0.358	0.379	0.385	0.376	0.376	0.361	0.352	0.339	0.315	0.305	0.305	(40)
(41)		taud	2.328	2.371	2.297	2.261	2.280	2.330	2.335	2.394	2.416	2.426	2.388	2.303	2.303	(41)
(42)		Ebn at Noon	215	249	263	271	275	278	276	275	263	242	215	195	195	(42)
(43)		Edh at Noon	20	25	33	38	40	38	37	33	29	24	19	17	17	(43)
(44)	All-Sky Solar Radiation	RadAvg	237	451	812	1254	1550	1669	1553	1279	964	542	284	182	182	(44)
(45)		RadStd	21	50	96	138	88	175	175	158	103	73	49	29	21	21

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page